

5. RESULT

5.1. SOCIO-DEMOGRAPHIC CHARACTERISTICS

A total of 3355 TB patient medical records in TB clinic of HIOWT FANA SPECIALIZED UNIVERSITY HOSPITAL from September 2006 to September 2008 E.C were registered. Of them a total of 384 sample were selected, among them 73 were reported TB and HIV co-infected, which indicates 19.0 % prevalence of HIV. Of whom, 37 (50.6%), were males. Out of 73 TB/HIV co-infected patient, 31 (42.7 %) were under age group of 30-44 years. Among 73 TB/HIV co-infected patients 42 (57.5 %) were married. (**table1**).

Table 1: Socio-demographic characteristics of HIV infected TB patients (N=73) in HIWOT FANA SPECIALIZED UNIVERSITY HOSPITAL from (September 2006 to September 2008 E.C).

Characteristics		Frequency	Percent
Sex	Male	37	50.7
	Female	36	49.3
age group	15-29	30	41.0
	30-44	31	42.7
	45-59	11	15.0
	≥60	1	1.3
marital status	Single	28	38.3
	Married	42	57.5
	Widowed	3	4.2
	Divorced	0	0
Educational status	Illiterate	15	20.5
	Read & write only	24	32.8
	Primary	5	6.8
	Secondary	4	5.4
	College	25	34.2
Place of residence	Urban	56	76.7
	Rural	17	23.3

5.2. CLINICAL CHARACTERISTICS

Sixty Nine (94.6 %) of the HIV infected TB patients were new cases and the rest, 2 (2.7 %) were relapse cases, 0 (0 %) were Rx after defaulting, 0 (0 %) were transfer in and 2 (2.7%) were others and 71(97.3%) patients had no history of TB treatment. All patients had no ART regimen changed during TB treatment. Out of the HIV infected TB patients, 12 (16.4%) had EPTB. Of them 4.1%, 0 % were TB capillary, TB peritonitis respectively and the rest 9 (12.3%) had unknown result. Looking into the distribution of types of TB among HIV infected TB patients; the proportion of SNPTB, SPPTB and EXPTB was 52.0 %, 31.5 % and 16.4 % respectively. Out of 73 TB/HIV co-infected patients recorded at TB clinic of HIWOT FANA SPECIALIZED UNIVERSITY HOSPITAL during the study period, 60 (15.6%) of the patients were WHO stage3 and 66 (17.2 %), 6 (1.6%) and 1 (0.3%) patients had CD4 count <350, 350-500, and >500 respectively during TB diagnosis. among 73 TB/ HIV co-infected patient 23 (31.5 %) were smear positive , 42 (57.5 %) smear negative and 8 (10.9 %) unknown smear status .(table2).

Table 2: Clinical characteristics of HIV infected TB patients (N=73) in HIWOT FANA SPECIALIZED UNIVERSITY HOSPITAL (September 2006 to September 2008 E.C).

Characteristics	Frequency	Percent
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Disease status	New	69	94.6
	Rx after Defaulted	0	0
	Relapse	2	2.7
	Transfer in	0	0
	Other	2	2.7
Types of TB	SPPTB	23	31.5
	SNPTB	38	52.0
	EXPTB	12	16.4
Smear status	Negative	42	57.5
	Positive	23	31.5
CD4 count during TB diagnosis group	<350	66	17.2
	350-500	6	1.6
	>500	1	1.0
WHO stage during TB Dx	Two	1	0.3
	Three	60	15.6
	Four	12	3.1

The overall treatment outcome of TB patients were 134(34.9 %) cured, 230 (59.9%) completed the treatment, 20 (5.2 %) Rx failure, 1 (0.2 %) died, and 3 (0.8 %) were transferred out. No treatment failure was recorded in HIV-positive TB patients (table3).

Table3: Treatment outcomes of TB patients (N=384) in HIWOT FANA SPECIALIZED UNIVERSITY HOSPITAL, (September 2006 to September 2008 E.C).

	TB treatment outcome	Total
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		Cured N (%)	Rx completed N (%)	Died N (%)	Rx failure N (%)	Transferred N (%)	N (%)
Sex	Male	68 (50.7)	126 (54.9)	1 (100)	11 (68.8)	2 (66.7)	
	Female	66 (49.3)	104 (45.3)	0 (0)	5 (31.2)	1 (33.3)	
age group	15-29	90 (67.1)	130 (56.5)	0 (0)	11 (68.7)	3 (100)	
	30-44	24 (17.9)	65 (28.2)	0 (0)	3 (18.7)	0 (0)	
	45-59	16 (12.6)	24 (10.4)	0 (0)	1 (6.2)	0 (0)	
	≥60	4 (2.9)	11 (4.7)	1 (100)	1 (6.2)	0 (0)	
marital status	Single	71 (52.9)	101 (43.9)	0 (0)	10 (62.5)	2 (66.6)	
	Married	62 (46.2)	121 (52.6)	0 (0)	6 (37.5)	1 (33.4)	
	Widowed	1 (0.7)	8 (3.4)	1 (100)	0 (0)	0 (0)	
	Divorced	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	
Educational level	Illiterate	36 (26.8)	57 (24.7)	0 (0)	2 (12.5)	3 (100)	
	Read and write only	18 (13.4)	41 (17.8)	0 (0)	4 (25)	0 (0)	
	Primary	15 (11.1)	30 (13.0)	0 (0)	3 (18.8)	0 (0)	
	Secondary	17 (12.7)	34 (14.8)	0 (0)	6 (37.5)	0 (0)	
	College	48 (35.8)	68 (29.6)	1 (100)	1 (6.3)	0 (0)	
TB classification	SPPTB	126 (94.0)	17 (7.4)	0 (0)	1 (6.3)	2 (66.7)	
	SNPTB	3 (2.3)	131 (56.9)	1 (100)	11 (68.7)	1 (33.3)	
	EPTB	5 (3.7)	82 (35.7)	0 (0)	4 (25)	0 (0)	
TB disease category	New	126 (94.0)	219 (95.2)	1 (100)	16 (100)	3 (100)	
	Rx after default	0 (0)	1 (0.43)	0 (0)	0 (0)	0 (0)	
	Relapse	8 (6.0)	0 (0)	0 (0)	0 (0)	0 (0)	
	Transfer in	0 (0)	1 (0.43)	0 (0)	0 (0)	0 (0)	
	Other	0 (0)	9 (3.9)	0 (0)	0 (0)	0 (0)	
history of TB treatment	Yes	8 (6.0)	0 (0)	0 (0)	0 (0)	0 (0)	
	No	126 (94.0)	230 (100)	1 (100)	16 (100)	3 (100)	
CD4 count during TB diagnosis	<350	17 (89.4)	47 (92.2)	0 (0)	1 (100)	1 (50.0)	
	350-500	1 (5.3)	4 (7.8)	0 (0)	0 (0)	1 (50.0)	
	>500	1 (5.3)	0 (0)	0 (0)	0 (0)	0 (0)	
WHO stage during TB Dx	Two	0 (0)	1 (2.0)	0 (0)	0 (0)	0 (0)	
	Three	16 (84.3)	41 (80.4)	0 (0)	1 (100)	2 (100)	
	Four	3 (15.7)	9 (17.6)	0 (0)	0 (0)	0 (0)	
Total							

SPPTB=Smear-Positive Pulmonary TB. SNPTB=Smear-Negative Pulmonary TB.
EXPTB=Extra pulmonary TB.

The overall treatment outcome of TB/HIV-infected patients were 19 (26.0 %) cured, 51 (69.9%) completed the treatment, 1 (1.4 %) Rx failure, 0 (0 %) died, and 2 (2.7 %) were transferred out. No treatment failure was recorded in HIV-positive TB patients (table4).

Table4: Treatment outcomes of TB/HIV co-infected patients (N=73) in HIWOT FANA SPECIALIZED UNIVERSITY HOSPITAL, (September 2006 to September 2008 E.C).

		TB /HIV co-infected treatment out come					
		Cured N (%)	Rx completed N (%)	Died N (%)	Rx failure N (%)	Transferr ed out N (%)	Total N (%)
Sex	male	8 (42.1)	27(52.9)	0 (0)	1(100)	1 (50.0)	37 (50.7
	female	11 (57.9)	24 (47.1)	0 (0)	0 (0)	1 (50.0)	36(49.3)
Age group	15-29	11(57.9)	17 (33.3)	0 (0)	0 (0)	2 (100)	30 (41.0)
	30-44	6 (31.6)	25 (49.0)	0 (0)	0 (0)	0 (0)	31(42.5)
	45-59	2 (10.5)	8 (15.7)	0 (0)	1 (100)	0 (0)	11 (15.1)
	>60	0 (0)	1 (1.1)	0 (0)	0 (0)	0 (0)	1 (1.4)
Marital status	married	9 (47.4)	31 (60.8)	0 (0)	1 (100)	1 (50.0)	42 (57.5)
	single	10 (52.6)	17 (33.3)	0 (0)	0 (00	1 (50.0)	28 (38.4)
	widowed	0 (0)	3 (5.9)	0 (0)	0 (0)	0 (0)	3 (4.1)
	divorced	0 (0)	0 (0)	0 (00	0 (0)	0 (0)	0 (0)
Education level	illiterate	4 (21.1)	9 (17.7)	0 (0)	0 (0)	2 (100)	15(20.5)
	Read and write only	5 (26.3)	18 (35.3)	0 (0)	1 (100)	0 (0)	24 (32.9)
	primary	1 (5.3)	4 (7.8)	0 (0)	0 (0)	0 (0)	5 (6.8)
	secondary	2 (10.5)	2 (3.9)	0 (0)	0 (0)	0 (0)	4 (5.5)
	Collage & above	7 (36.8)	18 (35.3)	0 (0)	0 (0)	0 (0)	25 (34.3)
Place of residence	urban	15 (79.0)	40(78.4)	0 (0)	1 (100)	0 (0)	56 (76.7)
	rural	4 (21.0)	11 (21.6)	0 (0)	0 (0)	2 (100)	17 (23.3)
TB disease	SPPTB	18 (94.7)	3 (5.9)	0 (0)	0 (0)	2 (100)	23 (31.5)

classification	SNPTB	1 (5.3)	37 (72.5)	0 (0)	0 (0)	0 (0)	38 (52.1)
	EXPTB	0 (0)	11 (21.6)	0 (0)	1 (100)	0 (0)	12 (16.4)
TB disease category	New	17 (89.5)	49 (96.0)	0 (0)	1 (100)	2 (100)	69 (94.6)
	Rx after default	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
	Relapse	2 (10.5)	0 (0)	0 (0)	0 (0)	0 (0)	2 (2.7)
	Transfer in	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
	Other	0 (0)	2 (4.0)	0 (0)	0 (0)	0 (0)	2 (2.7)
HX of TB Rx	Yes	2 (10.5)	0 (0)	0 (0)	0 (0)	0 (0)	2 (2.7)
	No	17 (89.5)	51 (100)	0 (0)	1 (100)	2 (100)	71 (97.3)
CD4 count during TB diagnosis	<350	17 (89.5)	47 (92.2)	0 (0)	1 (100)	1 (50.0)	66 (90.4)
	350-500	1 (5.3)	4 (7.9)	0 (0)	0 (0)	1 (50.0)	6 (8.2)
	>500	1 (5.3)	0 (0)	0 (0)	0 (0)	0 (0)	1 (1.4)
WHO staging during TB diagnosis	Two	0 (0)	1 (2.0)	0 (0)	0 (0)	0 (0)	1 (1.4)
	Three	16 (84.4)	41 (80.4)	0 (0)	1 (100)	2 (100)	60 (82.2)
	Four	3 (15.7)	9 (17.7)	0 (0)	0 (0)	0 (0)	12 (16.4)
Total		19	51	0	1	2	73 (100)

6. DISCUSSION

HIV co-infection among TB patients is well recognized as a major public health problem, especially in resource constraint settings. In this study, the prevalence of HIV co-infection among TB patients was 19.0 %; with a treatment success rate of 95.9 % which is higher than the studies conducted in Brazil (66 %) [32].

The observed difference of TB-HIV co-infection was partly due to local HIV prevalence difference in different parts of the country. In this study there were died HIV positive TB patients compared to study conducted in San-Francisco, United States of America, demonstrated that significantly more HIV co-infected patients died than other TB patients, and HIV co-infected TB patients that were on ART during the course of treatment for TB had a significantly lower mortality when compared with HIV infected patients who did not receive ART [35].

Unfavorable treatment outcomes in this study, including Rx failure 1 (1.4 %) and transferred out 2 (2.4 %) were lower than the findings in other studies.

The overall death rate was 1 (1.3 %) in this study, which is lower than that reported. A Study conducted in San-Francisco, United States of America, demonstrated that significantly more HIV co-infected patients died than other TB patient [35].

In this study TB treatment outcome between patients with TB/HIV co-infection and TB patients without HIV infection showed that, HIV co-infected TB patients have a higher treatment success rate (95.9 vs 94.5 %) and lower rate of treatment failure (1.4 vs 4.8 %) which is higher compared to a study conducted in Nigeria, with a treatment success rate of (48.8%v78.5%) and with a high rate of treatment failure (10.8%v4%).

Eighty six percent of the registered TB/HIV co-infection patients in this study were in the productive age group. This might indicate the negative impact of TB on the socioeconomic condition of the society. Since the study area is populated with substance using people this increases non stable sexually active people, thus study area signifies the urgent need urgent effective health intervention.

The greater proportion of SNPTB (52.0 %) cases observed among HIV infected TB patients in this study. This might be associated with variation in the concentration of acid fast bacilli in the sputum and the rate of caseation necrosis.

In this study the treatment success rate of women 35 (47.9 %) was equal to the men 35 (47.9 %).

The patients in the mean age greater than 30.55 years had lower treatment success rate compared to other age groups, older age has been reported to be a risk factor for death, partly due to co-infection and general physiological deterioration with age, and thus it is crucial to exercise close monitoring of TB treatment also in older patients [34]. Patients with SPPTB (9.6%) had higher treatment success rate compared to patients with EPTB (3.7%). This might be due to delayed diagnosis and treatment of EPTB patients, which I suspect increases tuberculosis mortality.

The success rate of all cases of TB treatment (29.8%) in this study was unsatisfactory when compared to the studies conducted in north India that reported a TSR of 77% [29] and that reported from studies done in southern Ethiopia (49%) [22] and Gambella Regional Hospital (63.4%) [27], but higher than that reported from studies done in Hareri Regional state (7%) [27]. TSR of 66.6% in HIV infected SPPTB patients was also reported in Nigeria study [30], which is higher than TSR reported in this study (9.6%). A greater success was reported in one study in Brazil [28], which observed a 55.8% cure rate among TB/ HIV co-infected patients. The low treatment success rate observed in this study might be attributed to the high transferred out rate (46.3%), where treatment outcome of patients who were transferred out were unknown. These findings are subject to at least two limitations. First, the hospital is a referral hospital and patients came from other health facilities to be diagnosed. Second, the patients were then admitted to the hospital and after finishing the intensive phase, transferred out to their nearest health facilities. Thus cured, failure, default and death were not detected.

